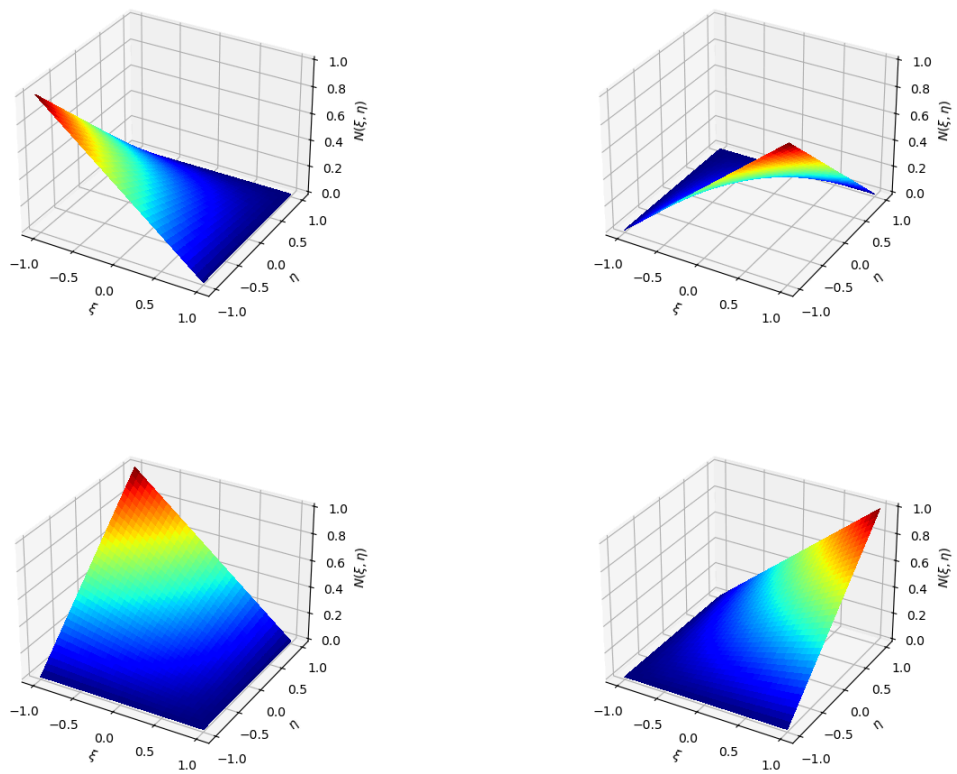
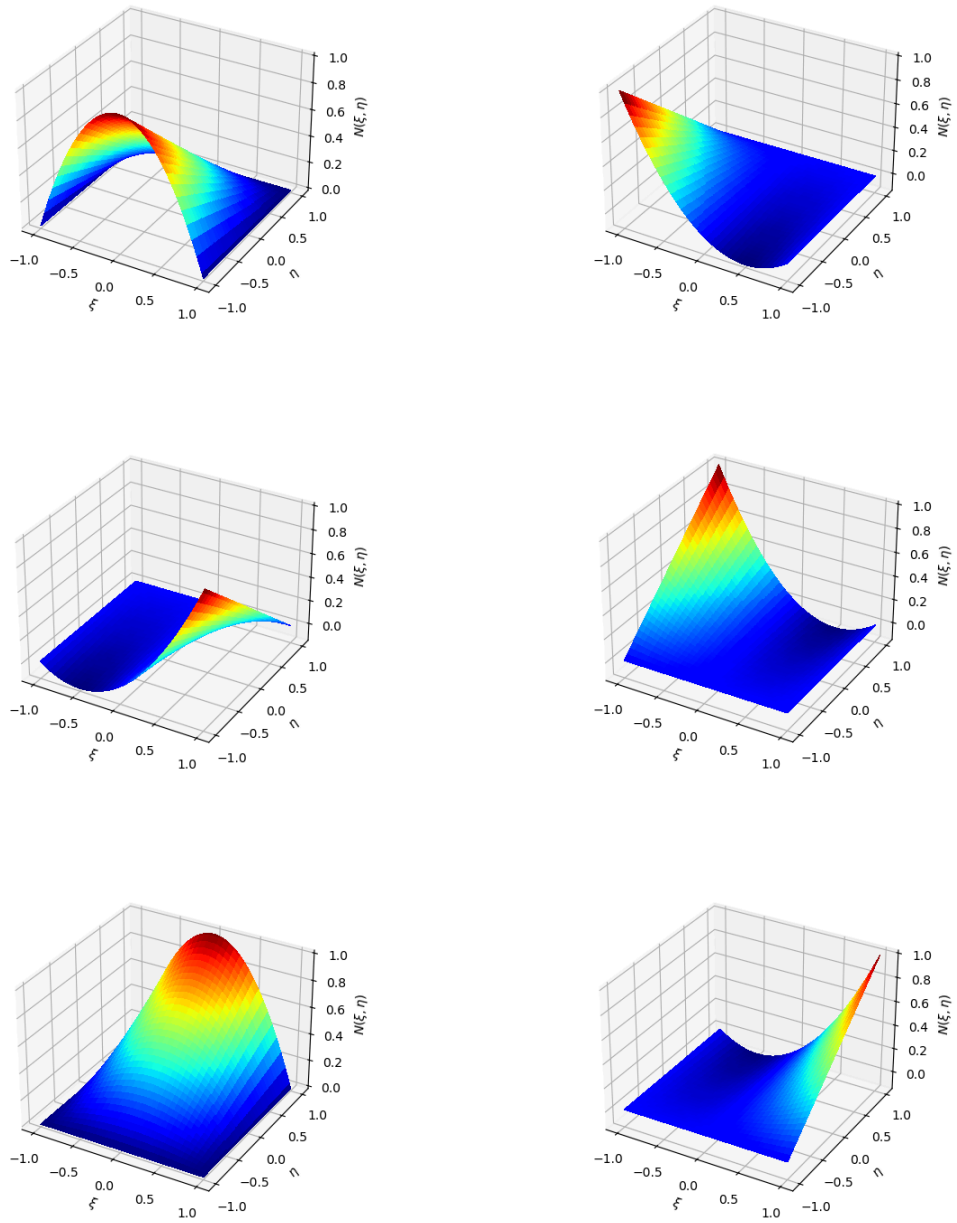
Problem 1

This problem involved finding 2-dimensional Lagrange polynomial basis functions, which are similar in some ways to their 1-dimensional counterparts. Below are the plots of all of the basis functions for the polynomial degrees in problem 1. The code used to generate these plots is included in the LearningSuite submission as multiDimensionalBasisFunction.py.

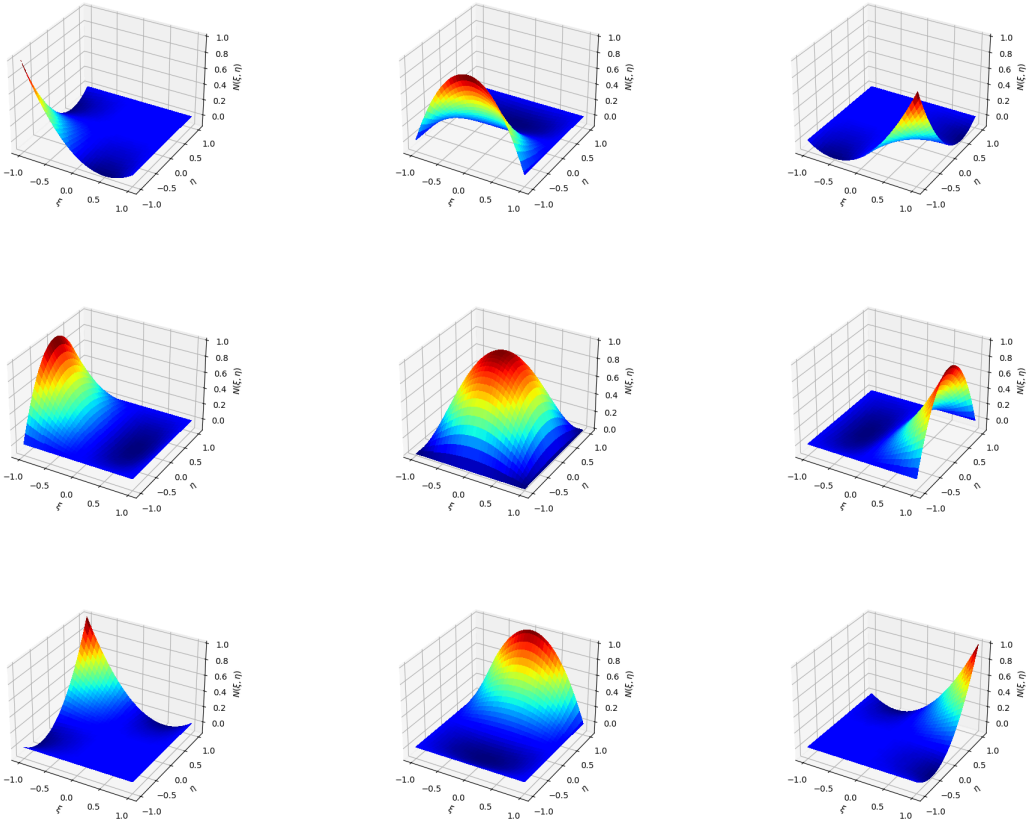
Polynomials of degree [1, 1]



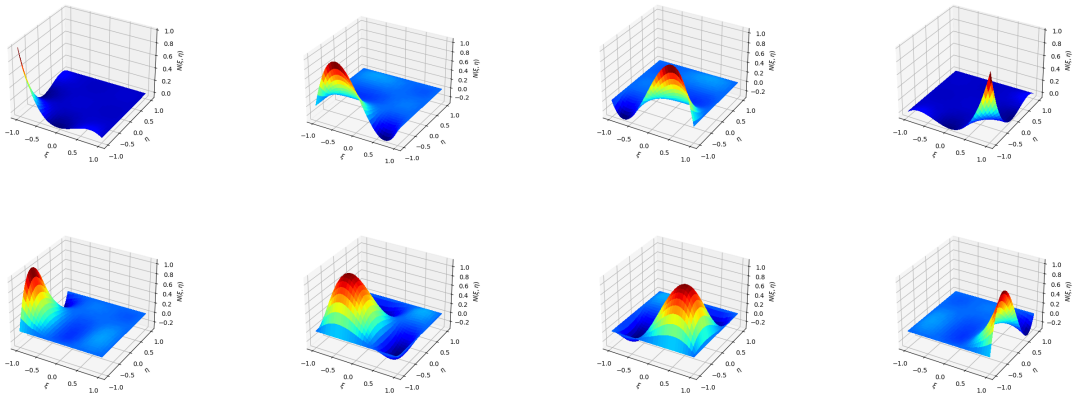
Polynomials of degree [2, 1]

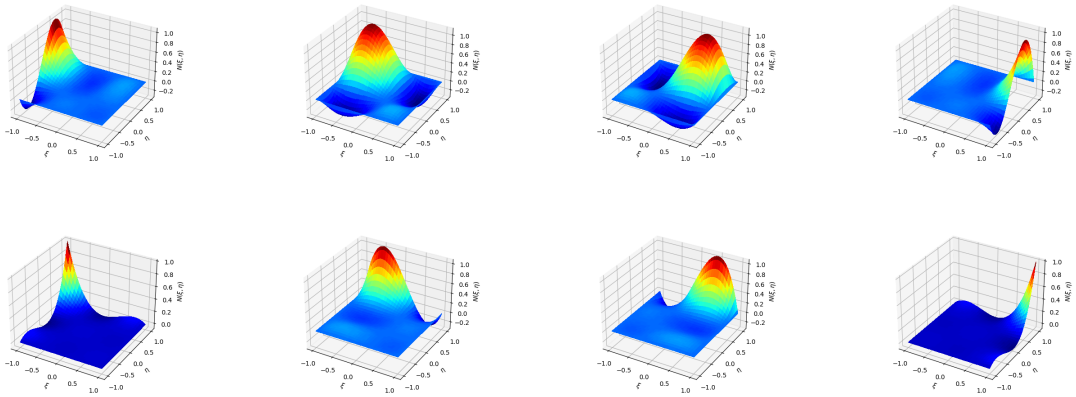


Polynomials of degree [2, 2]



Polynomials of degree [3, 3]

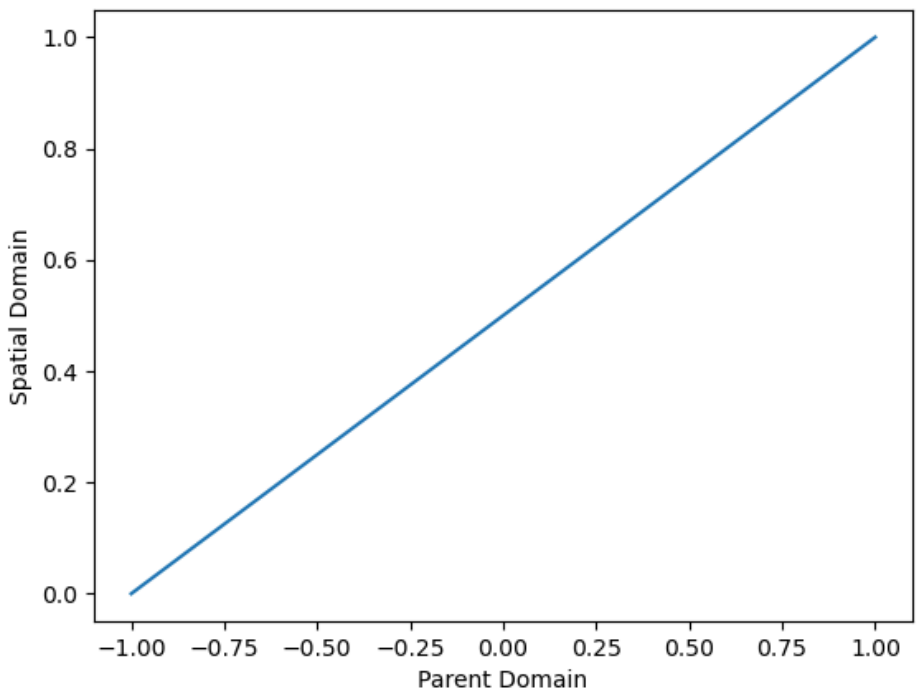




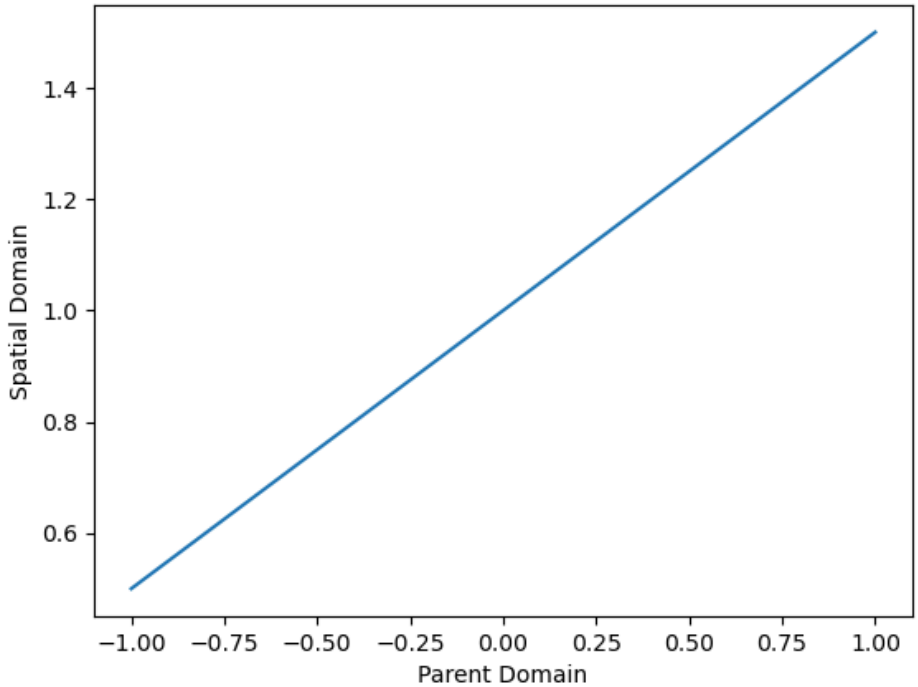
Problem 2

This problem involved using the Lagrange basis evaluations again to find a polynomial mapping from one domain to another given interpolation points in the two domains. Using the polynomial degrees and interpretation points from the problem statement, the plots below were generated. The code used to generate these plots is included in the LearningSuite submission as uniDimensionalXMap.py.

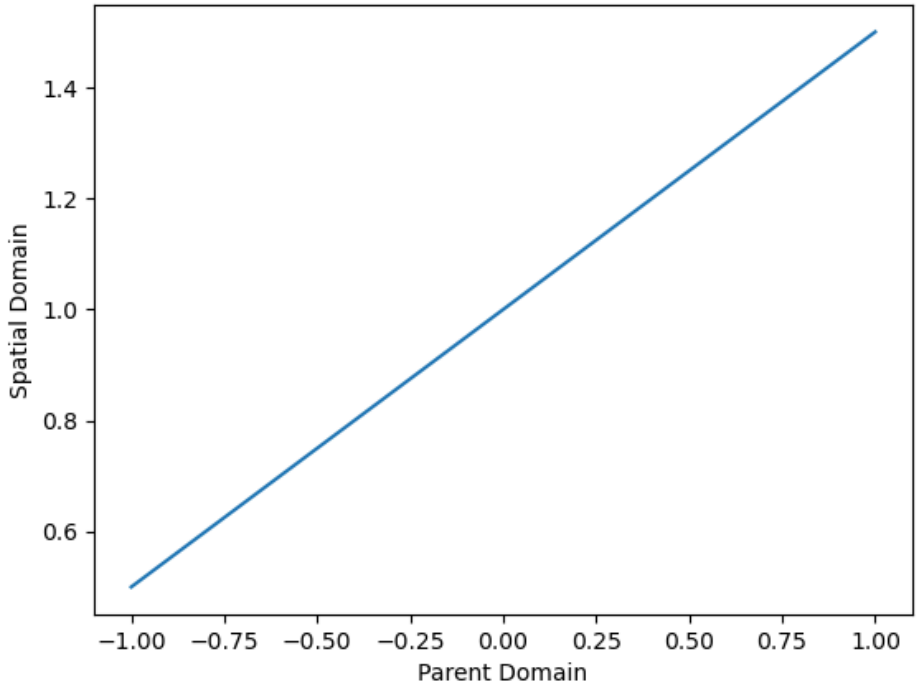
Degree 1 polynomial, spatial domain {0, 1}, parent domain {-1, 1}



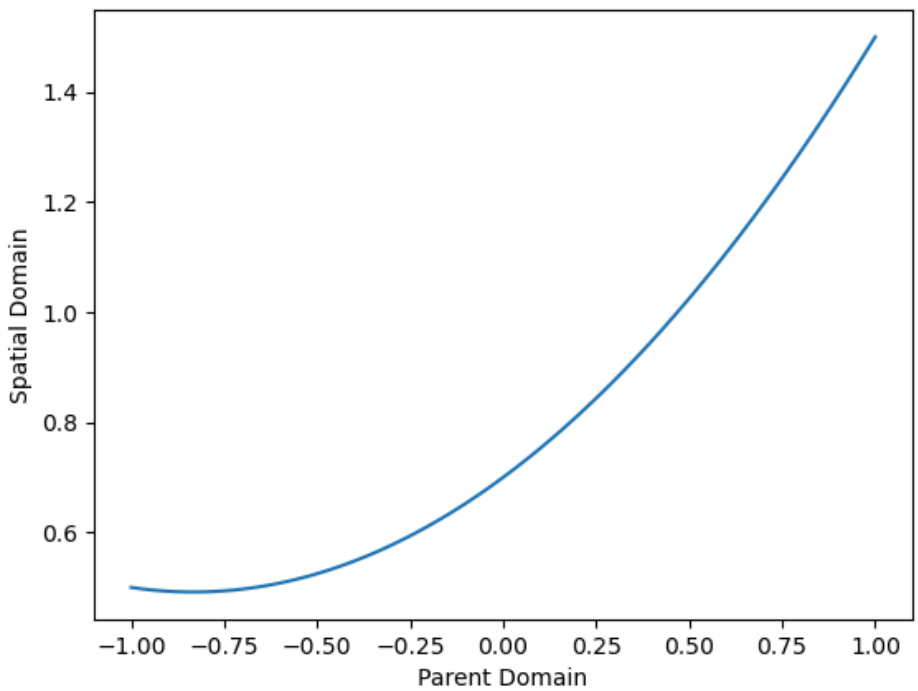
Degree 1 polynomial, spatial domain {0.5, 1, 1.5}, parent domain {-1, 0, 1}



Degree 1 polynomial, spatial domain {0.5, 0.7, 1.5}, parent domain {-1, -0.6, 1}



Degree 1 polynomial, spatial domain {0.5, 0.7, 1.5}, parent domain {-1, 0, 1}



Discussion Questions:

When is the mapping linear?
The mapping is linear when the points in the parent and spatial domain are evenly spaced out. In other words, if the three points given by (X_a, Xi_a) lie on the same line through the plane, then the mapping from the spatial domain to the parent domain will be linear. However, if they are not aligned or evenly spaced out, such as in the 4th example, the mapping will not be linear.

When does this map transition from the interval [-1, 1] to the input spatial element in a manner that correctly transforms points for Gaussian quadrature?
This happens when the mapping is linear and when the points along the parent domain are spaced out according to the Gaussian quadrature points needed.